Data Quality Audit Report

sample_sales

Generated: March 16, 2026 at 13:26

Data Health Score

Metric	Value
Total Rows	201
Total Columns	10
Health Score	74.9/100 (Good)
Critical Issues (High)	5
Warnings (Medium)	6
Info Items (Low)	3
Columns with Missing Data	4
Duplicate Rows	10

Executive Summary

This report presents a comprehensive data quality audit of **sample_sales**, containing **201 rows** and **10 columns**. The overall data health score is **74.9/100** — rated **Good**.

A total of **5 high-severity**, **6 medium-severity**, and **3 low-severity** issues were identified across 8 quality-check categories.

Key findings: **order_date** — 3 date format(s) detected: MM/DD/YYYY, DD-MM-YYYY, YYYY-MM-DD; **Unnamed: 7** — 103 missing values (51.2%); **revenue** — 43.3% of values stored as currency strings (e.g. "$1,200").

Immediate attention should be given to high-severity issues (missing values, duplicates, type inconsistencies) which have the greatest impact on analytical reliability. Detailed findings and prioritized recommendations follow.

Category	Weight	Score	Contribution
Completeness	30%	89.3	26.8
Uniqueness	15%	95.0	14.3
Type Consistency	20%	56.7	11.3
Outlier Reasonability	15%	99.3	14.9
Format Consistency	10%	25.0	2.5
Text Quality	10%	51.6	5.2
TOTAL	100%		74.9

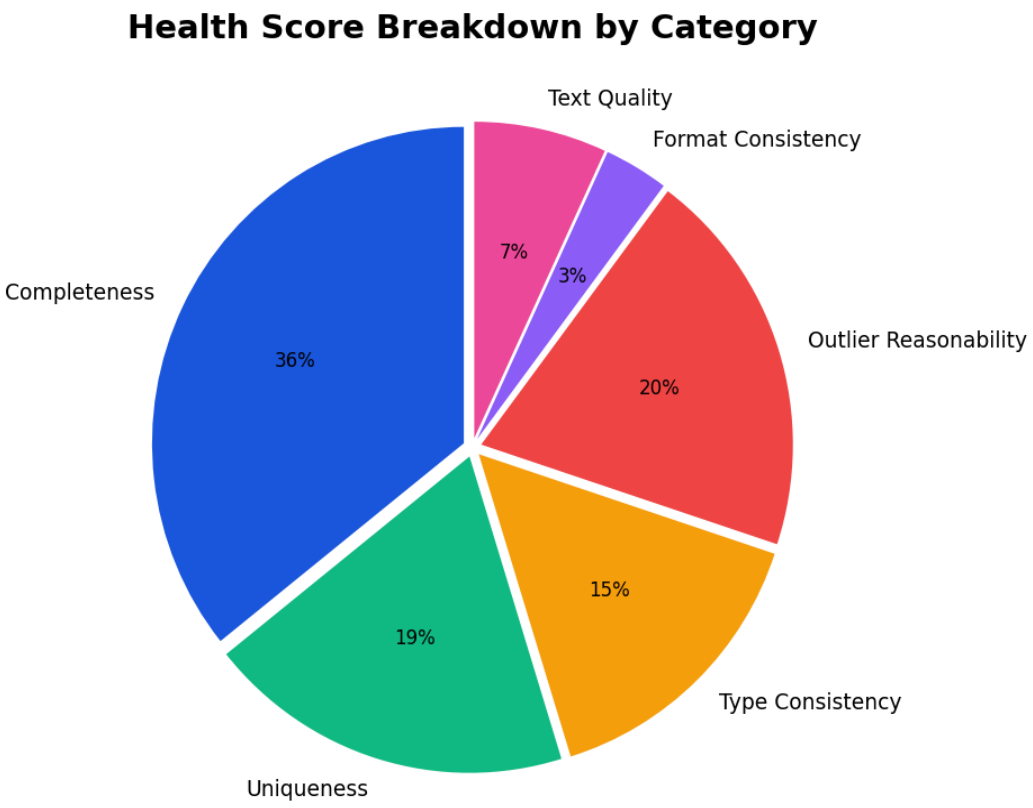

Figure 3: Health Score Breakdown by Quality Category

Missing Values Analysis

Of 10 columns, **4** contain missing values. Average missing rate: **10.7%**. Columns above 20% are flagged High and require imputation or removal.

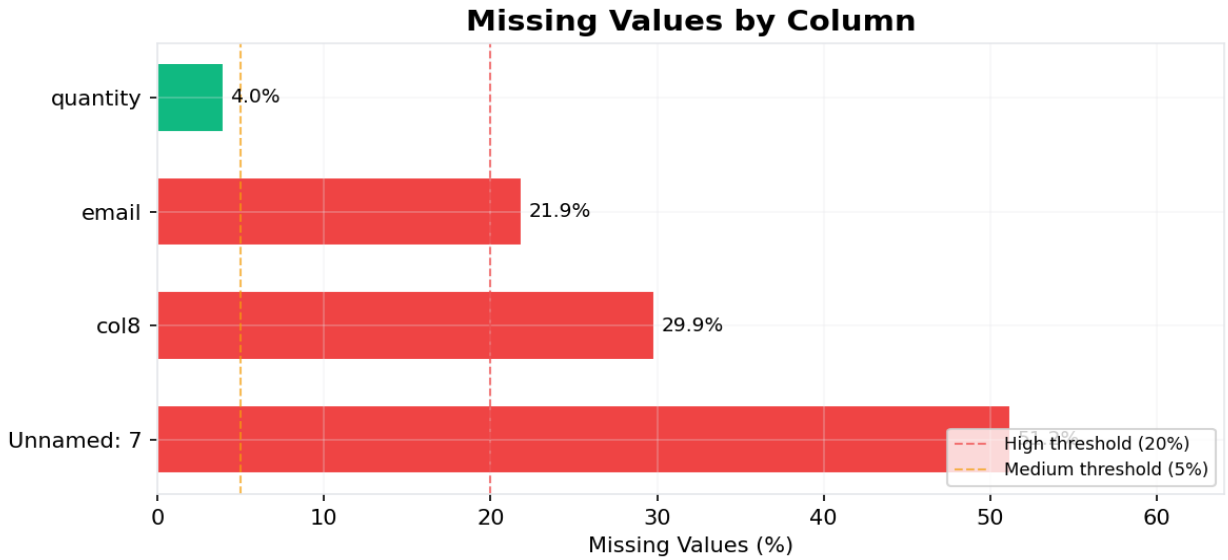

Figure 2: Missing values per column — red > 20%, orange 5–20%, green < 5%

Column	Missing Count	Missing %	Severity
Unnamed: 7	103	51.2%	High
col8	60	29.9%	High
email	44	21.9%	High
quantity	8	4.0%	Low

Issue Severity Overview

The heatmap maps each quality check (rows) against each column (columns). **H** = High severity, **M** = Medium, **L** = Low, blank = no issue.

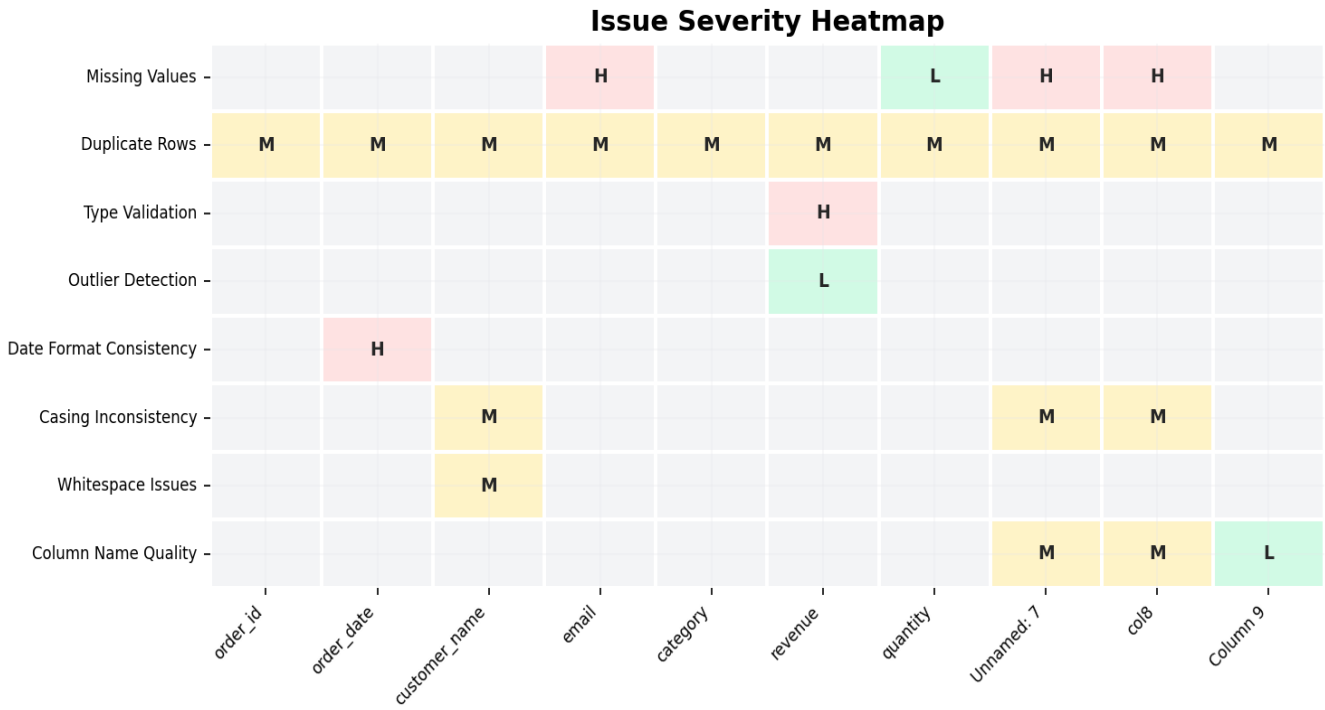

Figure 4: Issue Severity Heatmap

Top issue hotspots: **Unnamed: 7** (score 9), **col8** (score 9), **customer_name** (score 6), **revenue** (score 6), **email** (score 5).

Distribution Analysis

Histograms for 3 numeric column(s). Red dashed = median; orange dotted = IQR outlier boundaries (Q1 – 1.5×IQR, Q3 + 1.5×IQR). Values beyond orange lines are statistical outliers.

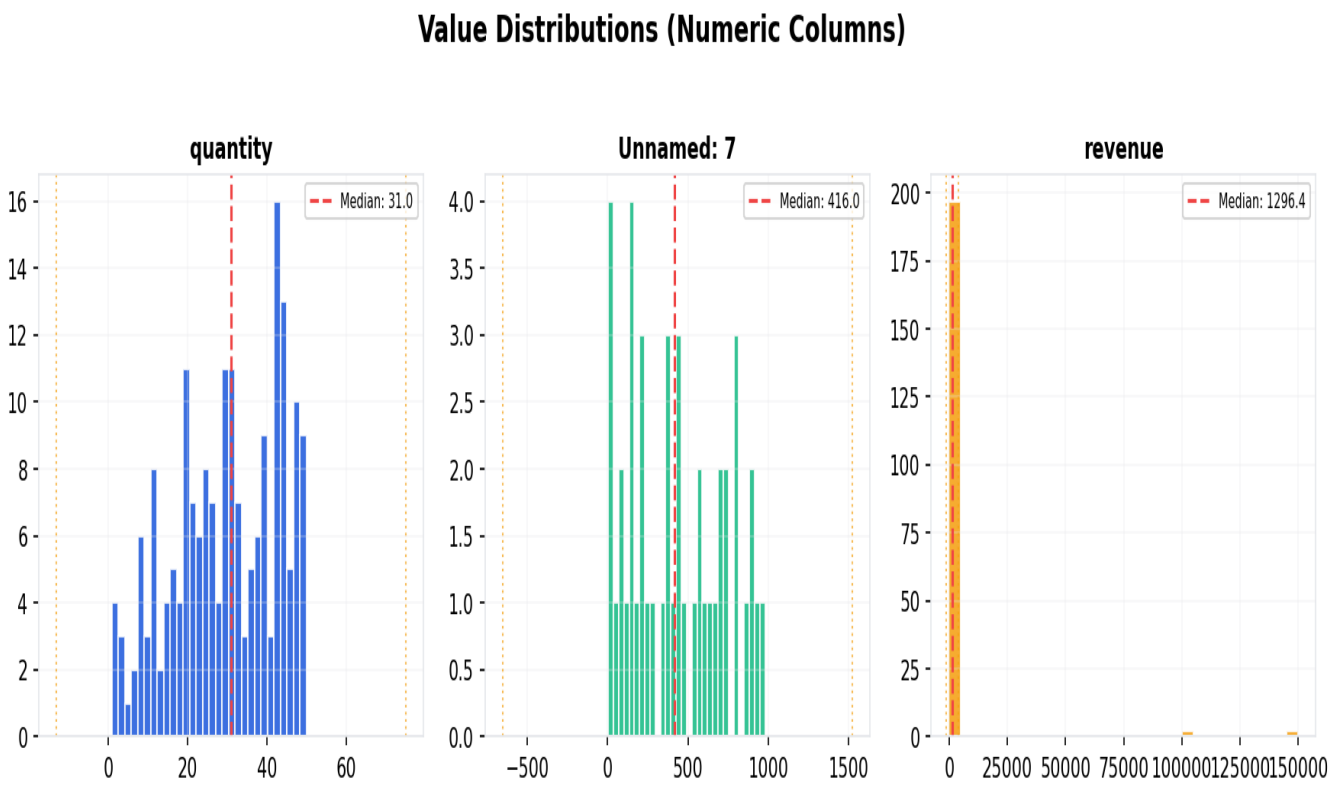

Figure 5: Value distributions for numeric columns

revenue: 4 outlier(s) (2.0%), valid range [-1304.41, 4024.95].

Trend Analysis

Numeric values over time using **order_date**. Raw data shown at low opacity; solid line = rolling average. Note: dates were parsed from mixed formats (MM/DD/YYYY, YYYY-MM-DD, DD-MM-YYYY).

Figure 6: Trends Over Time with rolling averages

Detailed Findings

All 14 issues sorted by severity (High → Low), then affected rows.

Column	Check Type	Severity	Finding	Rows	%
order_date	Date Format Consistenc	High	3 date format(s) detected: MM/DD/YYYY, DD-MM-YYYY, YYYY	201	100.0%
Unnamed: 7	Missing Values	High	103 missing values (51.2%)	103	51.2%
revenue	Type Validation	High	43.3% of values stored as currency strings (e.g. "$1,286	86	43.3%
col8	Missing Values	High	60 missing values (29.9%)	60	29.9%
email	Missing Values	High	44 missing values (21.9%)	44	21.9%
Unnamed: 7, col8	Column Name Quality	Medium	2 unnamed/auto-generated column(s)	201	100.0%
customer_name	Casing Inconsistency	Medium	Dominant casing covers only 36.8%: {'Title_Case': np.127	127	63.2%
col8	Casing Inconsistency	Medium	Dominant casing covers only 38.3%: {'Title_Case': np.87	87	61.7%
customer_name	Whitespace Issues	Medium	46 values with leading/trailing or double spaces (22.9%)	46	22.9%
Unnamed: 7	Casing Inconsistency	Medium	Dominant casing covers only 54.1%: {'ALL_CAPS': np.int6	45	45.9%
ALL	Duplicate Rows	Medium	10 exact duplicate rows (5.0%)	10	5.0%
Column 9	Column Name Quality	Low	1 column(s) with spaces/special chars in name	201	100.0%
quantity	Missing Values	Low	8 missing values (4.0%)	8	4.0%
revenue	Outlier Detection	Low	4 outliers (2.0%), valid range [-1304.41, 4024.95]	4	2.0%

Recommendations

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Priority	Issue	Recommendation	Affected Columns	Est. Impact
P1 — Critical	Missing values in Unnamed: 7	Impute (median/mode) or drop column if > 50% missing. 'Unnamed: 7' has 51.2% missing.	Unnamed: 7	+15.4 pts on Completeness
P1 — Critical	Mixed types in revenue	Strip '$' and ',' from revenue then cast to float. 43.3% of values stored as currency strings (e.g. "$1,200")	revenue	+8.7 pts on Type Consistency
P1 — Critical	Missing values in col8	Impute (median/mode) or drop column if > 50% missing. 'col8' has 29.9% missing.	col8	+9.0 pts on Completeness
P1 — Critical	Missing values in email	Impute (median/mode) or drop column if > 50% missing. 'email' has 21.9% missing.	email	+6.6 pts on Completeness
P2 — High	Unnamed / poorly-named columns	Rename or drop unnamed columns. Standardize all column names to snake_case. 2 unnamed/auto-generated column(s)	Unnamed: 7, col8	Improves maintainability
P2 — High	Inconsistent text casing	Apply .str.title() to name/category columns. Pick one convention and enforce it in the ingestion pipeline.	customer_name	+5 pts on Text Quality
P2 — High	Leading/trailing whitespace	Apply .str.strip() across all string columns. Use .str.replace(r'\s+', ' ', regex=True) for double spaces.	customer_name	+3 pts on Text Quality
P3 — Medium	Outliers in numeric columns (worst: revenue)	Investigate outlier rows — validate against source data. Use IQR capping (Winsorization) for extreme values. Do NOT blindly drop — confirm business context first.	quantity, Unnamed: 7, revenue	+0.1 pts on Outlier Reasonability